

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?
- The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{lcccccc} \widehat{WAGE} & = & -17.77 & + & 2.50EDUC & + & 0.23EXPER & + & 3.23METRO & - & 4.20FEMALE \\ (se) & & (2.36) & (0.14) & (0.031) & & (1.05) & & (0.81) & & \\ (robse) & & (2.50) & (0.16) & (0.029) & & (0.84) & & (0.80) & & \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

$$(a) \begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_1: \sigma_M^2 \neq \sigma_F^2 \end{cases}, N_m = 577$$

$$\hat{\sigma}_M^2 = \frac{97161.9174}{577-4} = 169.567$$

$$\hat{\sigma}_F^2 = 12.024^2 = 144.5766$$

$$GQ = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.567}{144.5766} \approx 1.173$$

$$F_{0.025}(573, 1000 - 577 - 4) = F_{0.025}(573, 419) \approx (0.838, 1.197)$$

$0.838 < 1.173 < 1.197 \rightarrow$ fail to reject H_0 , we don't have evidence to show that $\hat{\sigma}_M^2 \neq \hat{\sigma}_F^2$ at 95% level of significance.

$$(b) \begin{cases} H_0: \sigma_s^2 = \sigma_m^2 \\ H_1: \sigma_m^2 > \sigma_s^2 \end{cases}$$

$$\hat{\sigma}_s^2 = \frac{56231.0382}{400-5} \approx 142.36$$

$$\hat{\sigma}_m^2 = \frac{100703.0471}{600-5} \approx 169.25$$

$$\hat{F}_{0.95} = \frac{169.25}{142.36} \approx 1.189$$

$F_{0.95}(595, 395) \approx 1.105 < 1.189 \rightarrow \text{reject } H_0$, we have evidence to show that $\hat{\sigma}_m^2 > \hat{\sigma}_s^2$.

(c)

$$NR^2 = 59.03, K=5$$

$$59.03 \sim \chi^2_{(5-1)} = \chi^2(4)$$

$\chi^2(0.95, 4) \approx 9.488 < NR^2 = 59.03 \rightarrow \text{reject } H_0$, existing of heteroskedasticity at 95% we have evidence to show that the significant level.

$$(d) \chi^2(4) \approx 9.488 < 78.82 \rightarrow \text{reject } H_0$$

EDUC, EXPER, METRO, FEMACE,

EDUC², EXPER²,

EDUC x EXPER, EDUC x METRO, EDUC x FEMACE, EXPER x METRO,

EXPER x FEMACE, METRO x FEMACE

$$df = 4 + 2 + 6 = 12, \alpha = 0.05$$

$$\chi^2_{0.95, 12} \approx 21.026 < NR^2 = 78.82 \rightarrow \text{reject } H_0 \text{ at } 5\% \text{ critical value.}$$

(e)

EXPER, METRO, FEMACE have narrow interval estimates because their robust smaller the se.

intercept and EDUC have wider interval estimates

有不一致的现象, 因原本预期 robust 会较大, 其有考虑异质变异性

(f)

compatible

$\therefore$ (b) 做了异质性 test 且存在异质性,

而 (f) 用了 robust se 来估计不显著,

$\therefore$ 合理,

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

(a)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

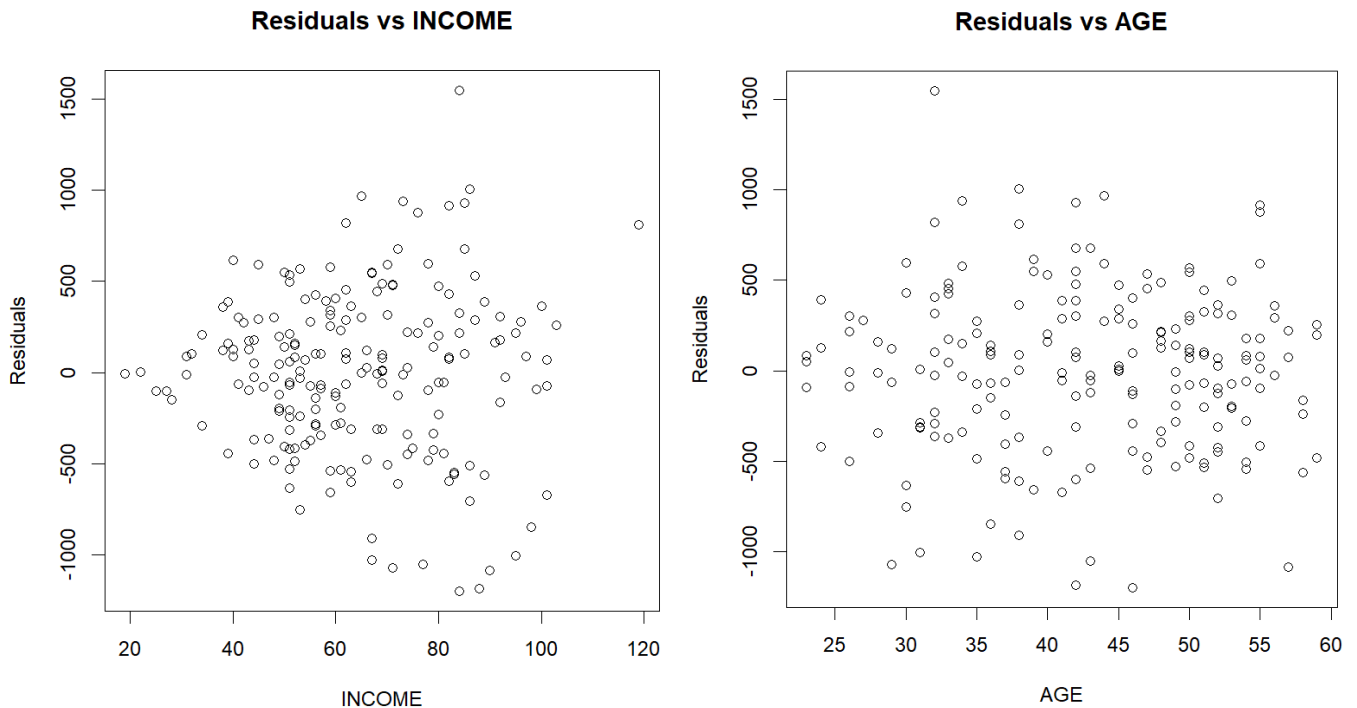
Residuals:
    Min       1Q   Median       3Q      Max
-1198.14  -295.31   17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775  -2.306   0.0221 *
income       14.201      1.800    7.889 2.10e-13 ***
age          15.741      3.757    4.189 4.23e-05 ***
kids        -81.826     27.130   -3.016  0.0029 **
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,    Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16

>
> confint(ols_model, "kids", level = 0.95)
           2.5 %    97.5 %
kids -135.3298 -28.32302
```

(b)



income $\uparrow$, residuals $\uparrow \Rightarrow$ 可能存在異質變異性

(c)

```
> cat("GQ statistic:", round(gq_stat, 3), "\n")
GQ statistic: 3.104
> cat("Critical value at 5% level:", round(f_crit, 3), "\n")
Critical value at 5% level: 1.429
>
> if (gq_stat > f_crit) {
+   cat("Reject H0: Evidence of heteroskedasticity.\n")
+ } else {
+   cat("Fail to reject H0: No evidence of heteroskedasticity.\n")
+ }
Reject H0: Evidence of heteroskedasticity.
```

(d)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-391.5480	142.6548	-2.7447	0.0066190
income	14.2013	1.9389	7.3246	6.083e-12
age	15.7409	3.9657	3.9692	0.0001011
kids	-81.8264	29.1544	-2.8067	0.0055112

```
(Intercept) **
income      ***
age         ***
kids        **
---
```

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> cat("\n95% robust confidence interval for 'kids':\n")
```

```
95% robust confidence interval for 'kids':
> cat("(", round(lower, 4), ",", round(upper, 4), ")\n")
( -138.969 , -24.6838 )
```

(e)

```
> cat("GLS (conventional SE) 95% CI for kids:\n")
GLS (conventional SE) 95% CI for kids:
> cat("(", round(ci_gls_conventional[1], 4), ",", round(ci_gls_conventional[2], 4), ")\n\n")
( -119.8945 , -33.7181 )

>
> vcov_gls_robust <- vcovHC(gls_model, type = "HC1")
> coef_kids_gls <- coef(gls_model)["kids"]
> se_kids_gls_robust <- sqrt(vcov_gls_robust["kids", "kids"])
>
> lower_gls_robust <- coef_kids_gls - 1.96 * se_kids_gls_robust
> upper_gls_robust <- coef_kids_gls + 1.96 * se_kids_gls_robust
>
> cat("GLS (robust SE) 95% CI for kids:\n")
GLS (robust SE) 95% CI for kids:
> cat("(", round(lower_gls_robust, 4), ",", round(upper_gls_robust, 4), ")\n\n")
( -121.1388 , -32.4738 )
```

相較 (a), (d) 有更窄的 C.I